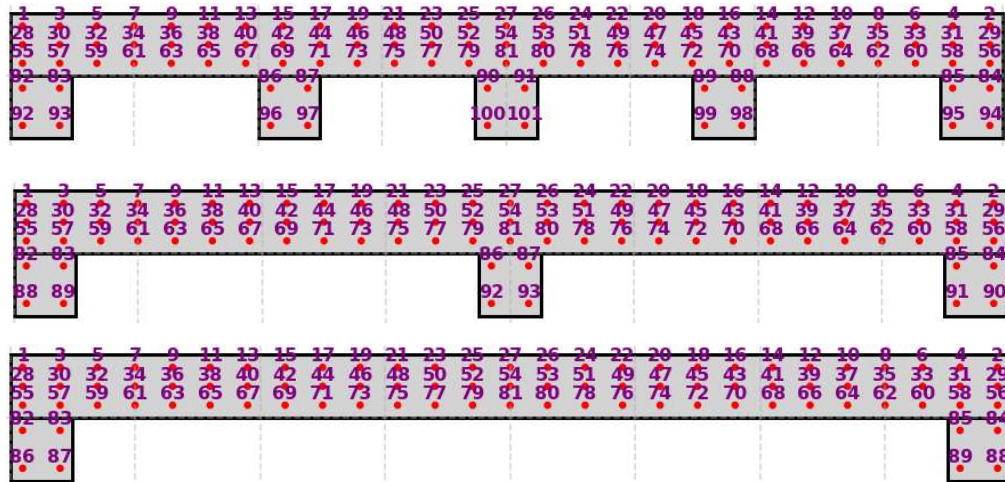
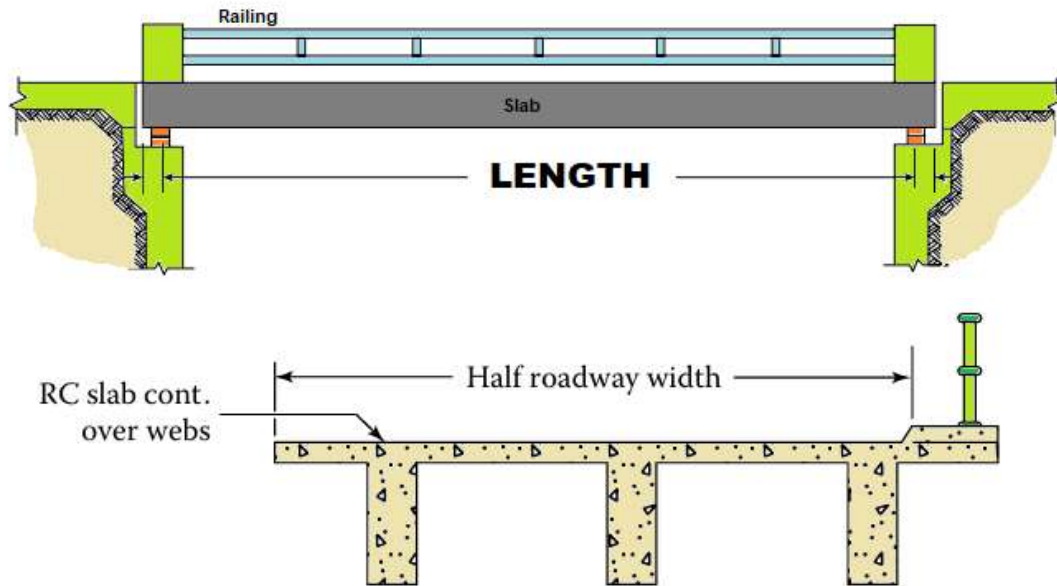
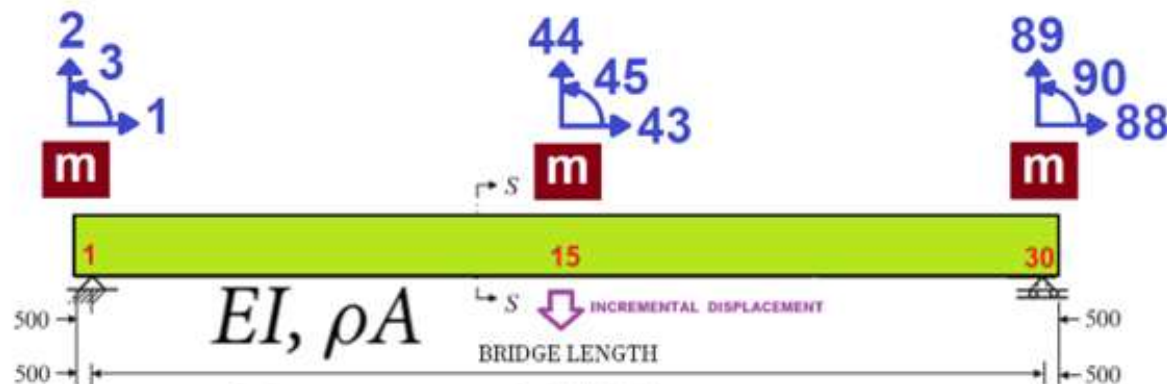


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A CONCRETE T SECTION SLAB SUPERSTRUCTURE SIMPLY SUPPORTED SHORT SPAN BRIDGE: AN OPENSEES FRAMEWORK FOR MOMENT-ROTATION ANALYSIS

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)





$$P(t) = P_0 e^{-0.05 \bar{\omega} t} \sin(\bar{\omega} t)$$

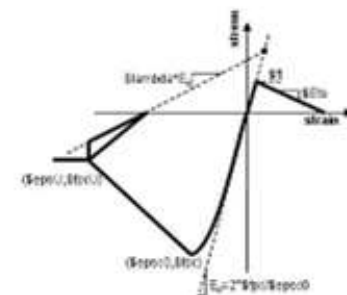


$$\text{Structure Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

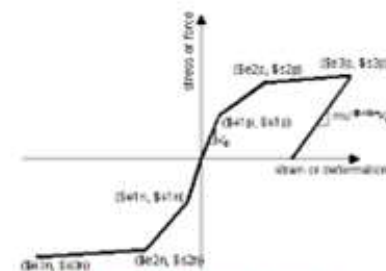
Δ_d = Lateral Displaement from Dynamic Analysis

Δ_y = Lateral Yield Displaement from Pushover Analysis

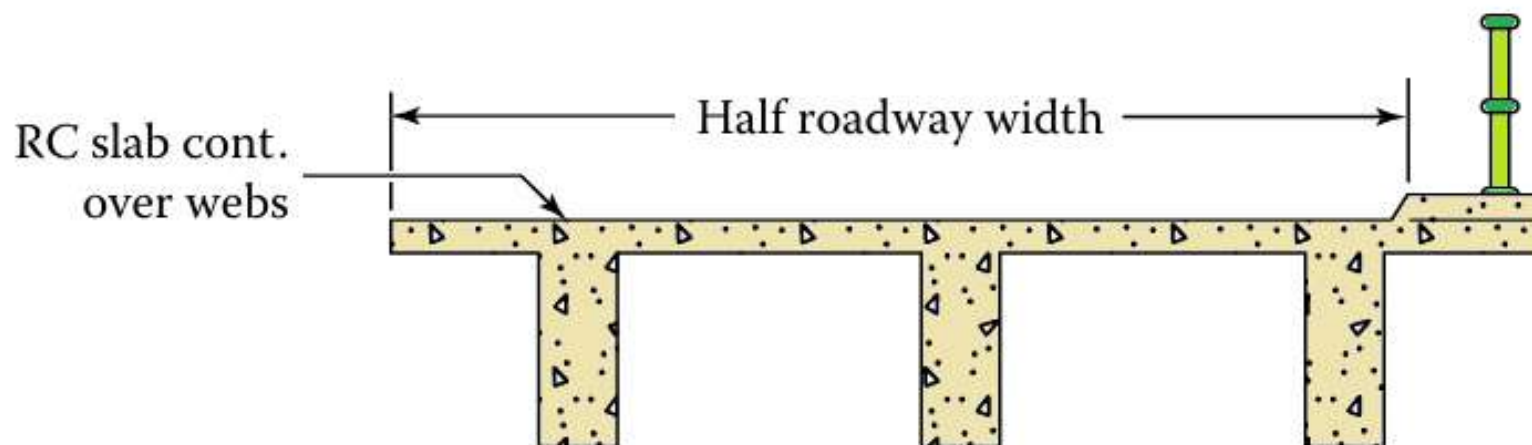
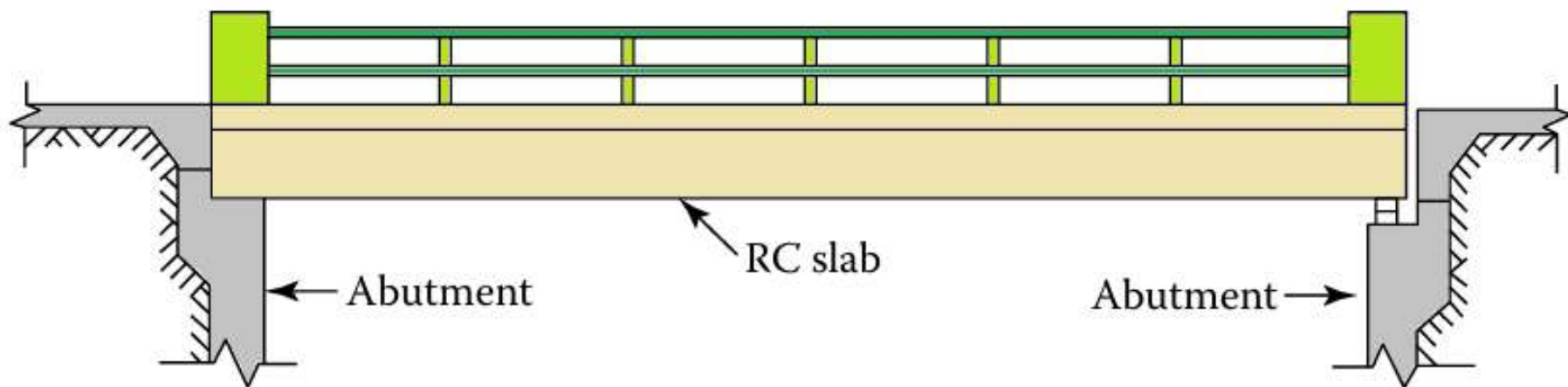
Δ_u = Lateral Ultimate Displaement from Pushover Analysis



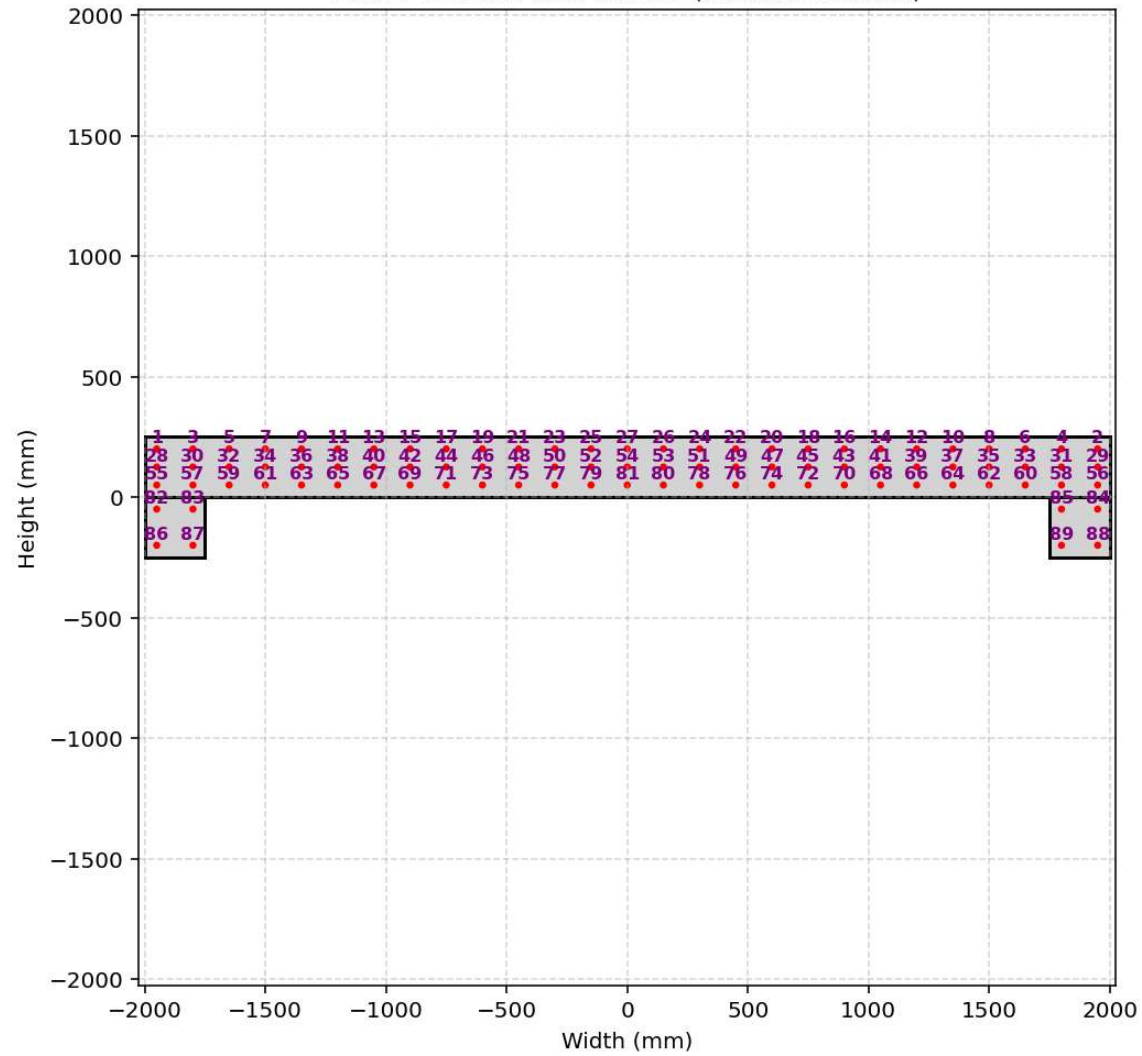
CORE AND COVER CONCRETE RELATION



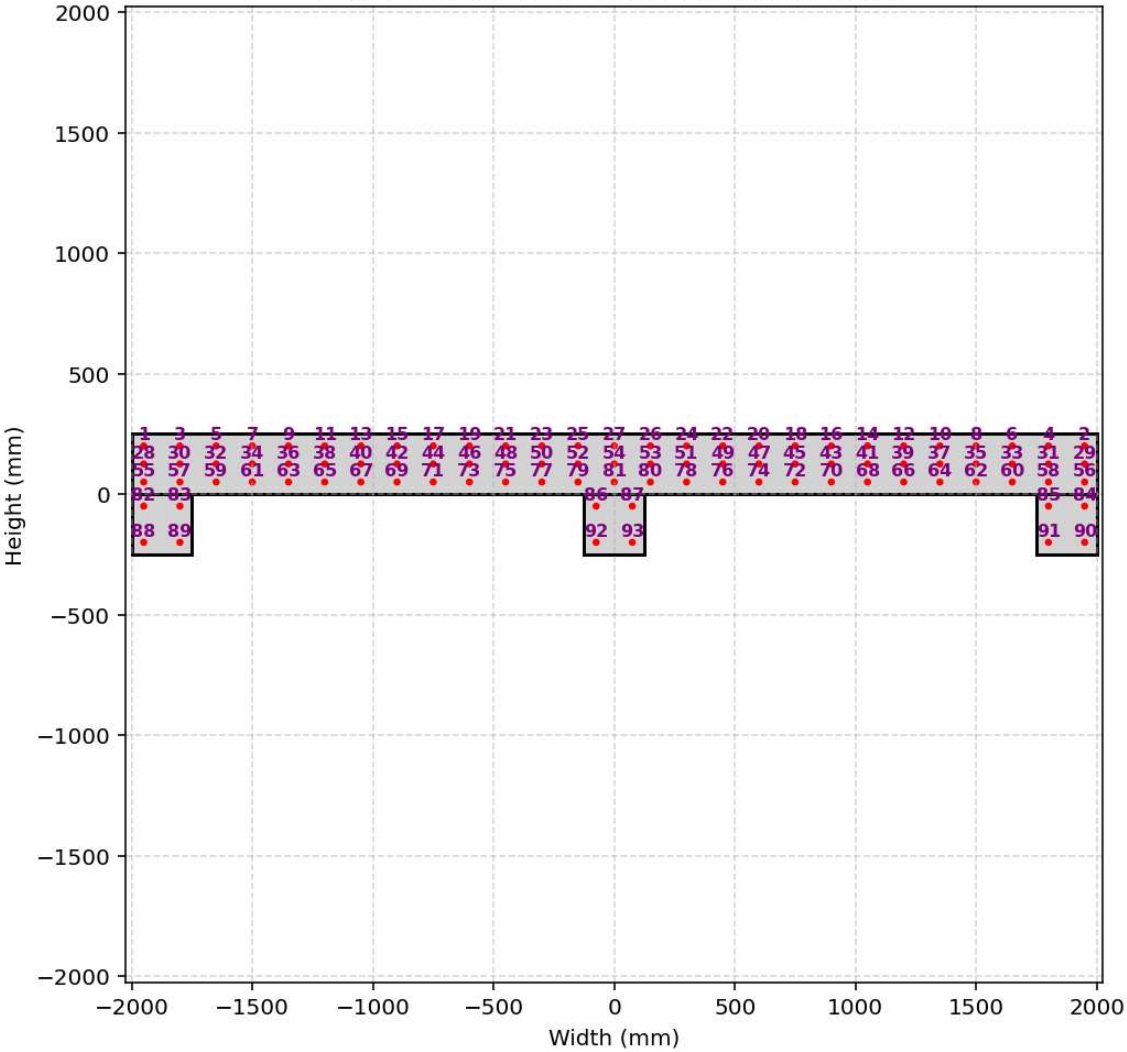
STEEL REBAR SECTION RELATION



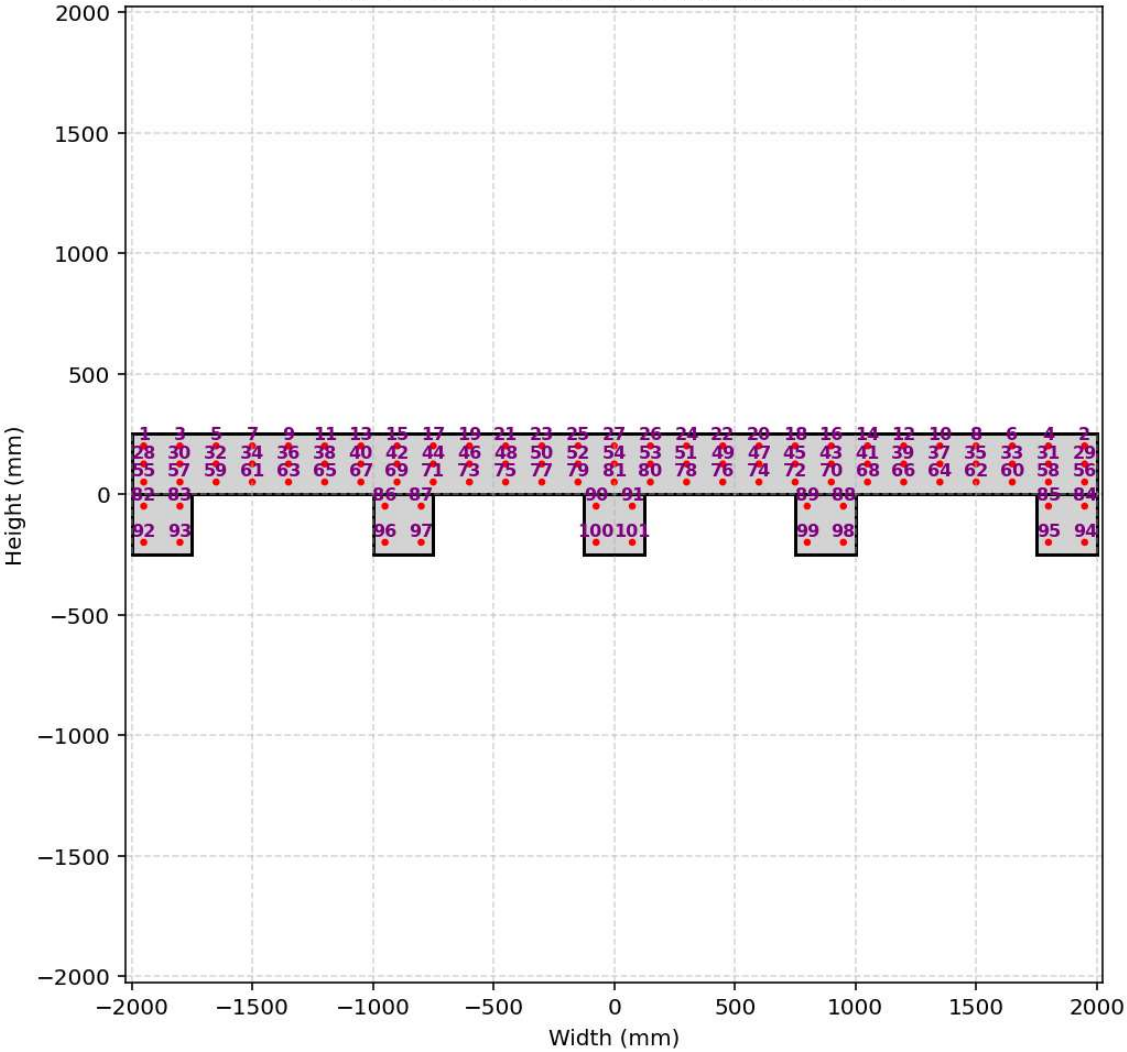
Slab T Section with Rebars (numbers shown)



Slab T Section with Rebars (numbers shown)



Slab T Section with Rebars (numbers shown)



Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\ DELL\Desktop\OPENSEES_FILES\+EIGHT_ANALYSIS\BRIDGE_T_SECTION_MOMENT-ROTATION_ANALYSIS.py

BRIDGE_T_SECTION_M...TATION_ANALYSIS.py x CONCRETE_SLAB_T_SECTION_FUN_EXTRA.py x

```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A CONCRETE T SECTION SLAB SUPERSTRUCTUR
4 # SUPPORTED SHORT SPAN BRIDGE : AN OPENSEES FRAMEWORK FOR MOMENT-ROTATION ANALYSIS
5 #-----
6 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EV
7 # OF ENERGY DISSIPATION CAPACITY INDEX
8 #-----
9 # THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
10 # EMAIL: salar.d.ghashghaei@gmail.com
11 #####
12 """
13 Nonlinear Seismic Performance Assessment of a simply supported concrete slab short span br
14 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and E
15
16 This OpenSeesPy script performs rigorous nonlinear static analysis of a simply supported c
17 for performance-based earthquake engineering. The 2D model incorporates both material nonl
18 (elastic-perfectly plastic or hysteretic steel with strain hardening)
19 and geometric nonlinearity to capture P-delta effects
20 and large displacements-essential for collapse assessment.
21
22 Eight analysis protocols are implemented:
23 (1) [PERIOD] : Structural Period
24 (2) [STATIC] : Gravity load analysis establishing dead load state
25 (3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
26 and plastic mechanism identification
27
28 The code continuously records element axial forces, stresses, strains,
29 and full-field nodal displacements, enabling member-level demand-to-capacity
30 ratio tracking. Period monitoring during inelastic response reveals softening
31 and potential period shift into resonant ground motion frequency ranges.
32 This framework enables vulnerability curve development, seismic fragility assessment,
33 and retrofit prioritization-directly supporting next-generation bridge seismic design code
34 and risk-informed asset management decisions.
```

Period of Structure During Pushover Analysis

Structural Period [s]

Displacement [mm]

PERIOD - MIN VALUES: Min: 0.087 (s) - Mean: 0.122 (s) - Max: 0.148 (s)

PERIOD - MAX VALUES: Min: 0.323 (s) - Mean: 0.476 (s) - Max: 0.620 (s)

[Python Console] Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 4, Col 61 UTF-8 CRLF RW Mem 43%

STATIC ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\HEIGHT_ANALY...ONCRETE_SLAB_T_SUPERSTRUCTURE_SIMPLY_SUPPORTED.py

P(t)_BRIDGE_CONCRE...IMPLY_SUPPORTED.py x CONCRETE_SLAB_T_SECTION_FUN_EXTRA.py x

```
440 # Compute modal properties
441 ops.modalProperties("-print", "-file", f"SALAR_ModalReport_{ANAL_TYPE}.txt", "-unorm
442 return PERIODmin, PERIODmax
443
444 if ANAL_TYPE == 'STATIC':
445     ops.integrator('LoadControl', 1.0)
446     ops.analysis('Static')
447     OK = ops.analyze(1)
448     S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
449     ops.reactions()
450     reaction.append(ops.nodeReaction(1, 2)+ops.nodeReaction(30, 2)) # SHEAR BASE REACT
451     disp_mid.append(ops.nodeDisp(center_node, 2)) # DISPLACEMENT NOD
452     dispX.append(ops.nodeDisp(center_node, 1)) # DISPLACEMENT NOD
453     dispY.append(ops.nodeDisp(center_node, 2)) # DISPLACEMENT NOD
454     # Store axial forces, strain and stress
455     for ele_id in ele_axialforce.keys():
456         ele_axialforce[ele_id].append(ops.eleResponse(ele_id, 'force')[0]) # [N
457         ele_shearforce[ele_id].append(ops.eleResponse(ele_id, 'force')[1]) # [N
458         ele_momentforce[ele_id].append(ops.eleResponse(ele_id, 'force')[2]) # [N
459     # Store displacements
460     for node_id in node_displacementsX.keys():
461         node_displacementsX[node_id].append(ops.nodeDisp(node_id, 1))
462         node_displacementsY[node_id].append(ops.nodeDisp(node_id, 2))
463     else:
464         print('\n\nSTATIC ANALYSIS DONE.\n\n')
465
466     DATA = (reaction, disp_mid,
467             ele_axialforce, ele_shearforce, ele_momentforce,
468             node_displacementsX, node_displacementsY,
469             dispX, dispY)
470
471     return DATA
472
473 if ANAL_TYPE == 'PUSHOVER': # STATIC TIME-HISTORY ANALYSIS
```

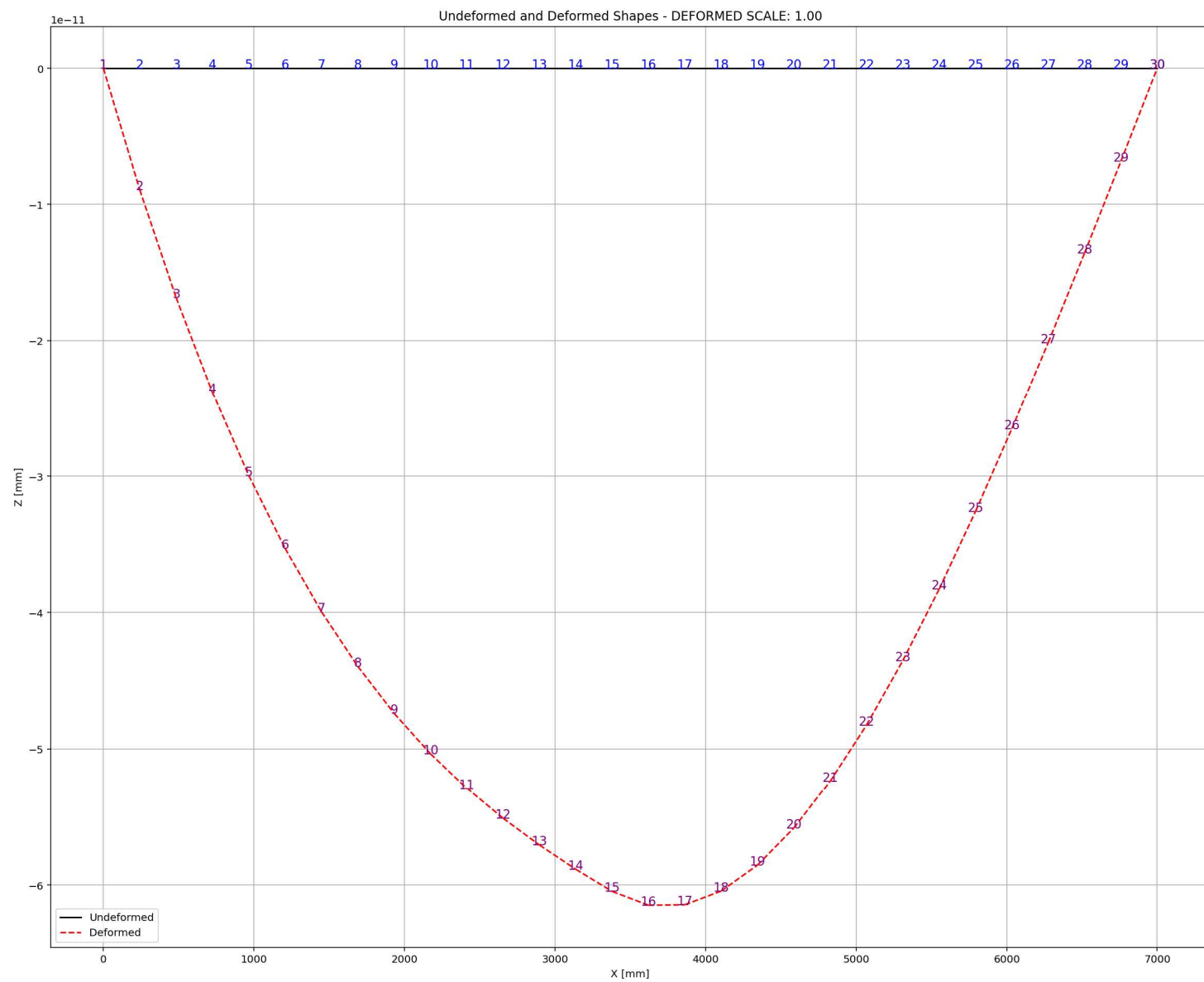
Slab T Section with Rebars (numbers shown)

Height (mm)

Width (mm)

IPython Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 87, Col 24 UTF-8 CRLF RW Mem 46%



PUSHOVER ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\De\l\Desktop\OPENSEES_FILES\+EIGHT_ANALY...YSIS\BRIDGE_T_SECTION_MOMENT-ROTATION_ANALYSIS.py

BRIDGE_T_SECTION_M...TATION_ANALYSIS.py x CONCRETE_SLAB_T_SECTION_FUN_EXTRA.py x

```
78
79 # BOOK: Highway Bridge Superstructure Engineering; LRFD Approaches to Design and Analysis -
80 #%%-----
81 import numpy as np
82 import openseespy.opensees as ops
83 import matplotlib.pyplot as plt
84 import PLOT_2D as S01
85 import ANALYSIS_FUNCTION as S02
86 import PERIOD_FUN as S03
87 import DAMPING_RATIO_FUN as S04
88 import EIGENVALUE_ANALYSIS_FUN as S05
89 import RAYLEIGH_DAMPING_FUN as S06
90 import BILINEAR_CURVE as S07
91 import SECTION_CURVATURE_FUN as S09
92 import SECTION_STRESS_STRAIN_FUN as S10
93 import CONCRETE_SLAB_T_SECTION_FUN_EXTRA as S14
94 import CONCRETE_SLAB_U_SECTION_FUN_EXTRA as S16
95 import CONCRETE_SLAB_E_SECTION_FUN_EXTRA as S17
96 #%%-----
97 def SHORT_SPAN_T_SLAB_CONCRETE_BRIDGE(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS):
98     # Initialize model
99     ops.wipe()
100     ops.model('basic', '-ndm', 2, '-ndf', 3)
101
102     MAX_ITERATIONS = 5000 # Maximum number of iterations
103     MAX_TOLERANCE = 1.0e-10 # Specified tolerance for convergence
104
105     #%% GEOMETRY DEFINITION
106     NUM_POINTS = 30 # Structural Node counts
107     e = 0.001 * LENGTH # [mm] Bridge deck camber (Imperfection amplitude)
108     # Camber is the slight upward curvature intentionally built into a bridge deck so that
109     dy = LENGTH / (NUM_POINTS-1) # Element length
110     for i in range(NUM_POINTS):
111         x = i * dy
```

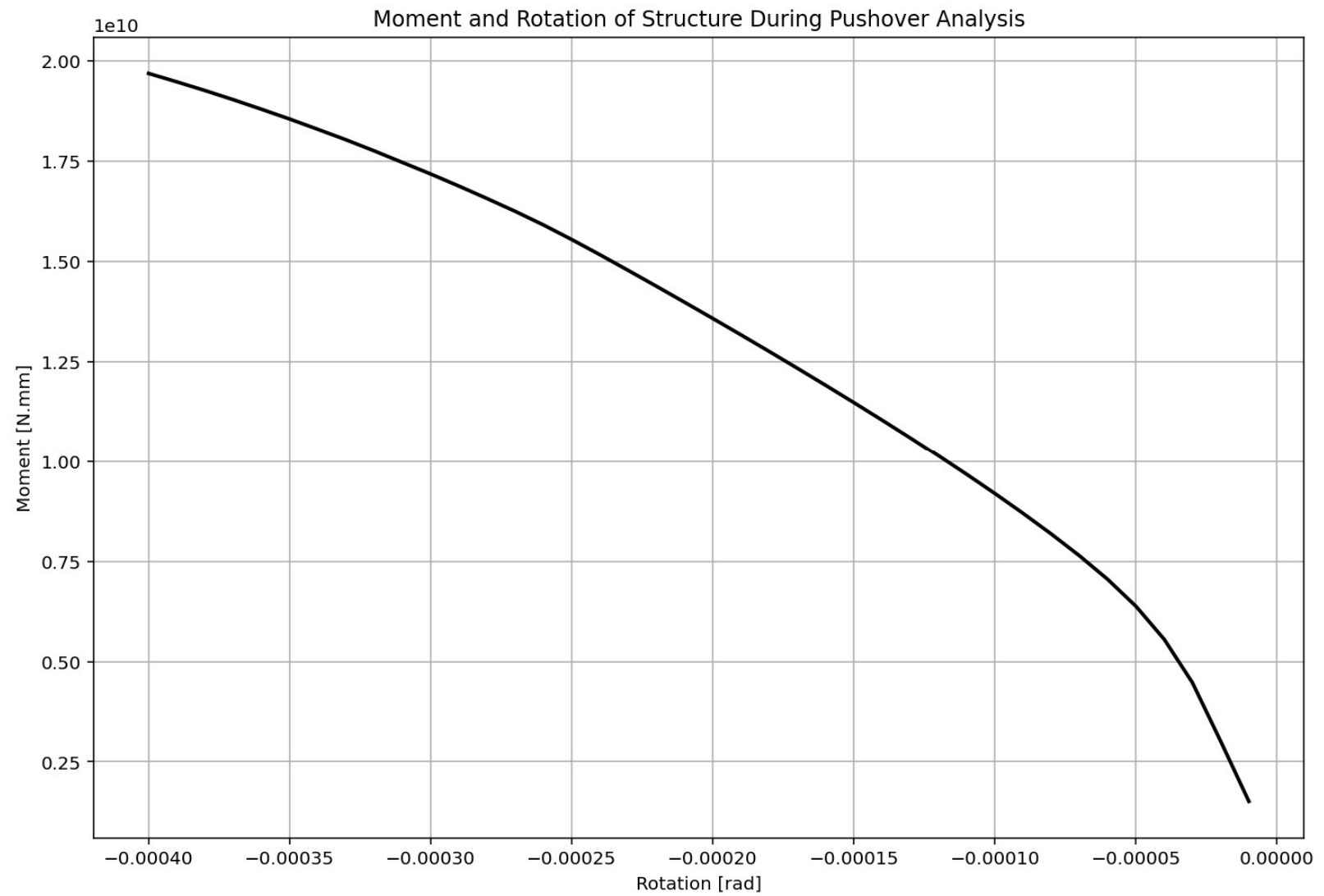
..E_SIMPLY_SUPPORTED_8_ANA\SECTION_MOMENT-ROTATION_ANALYSIS

30 %

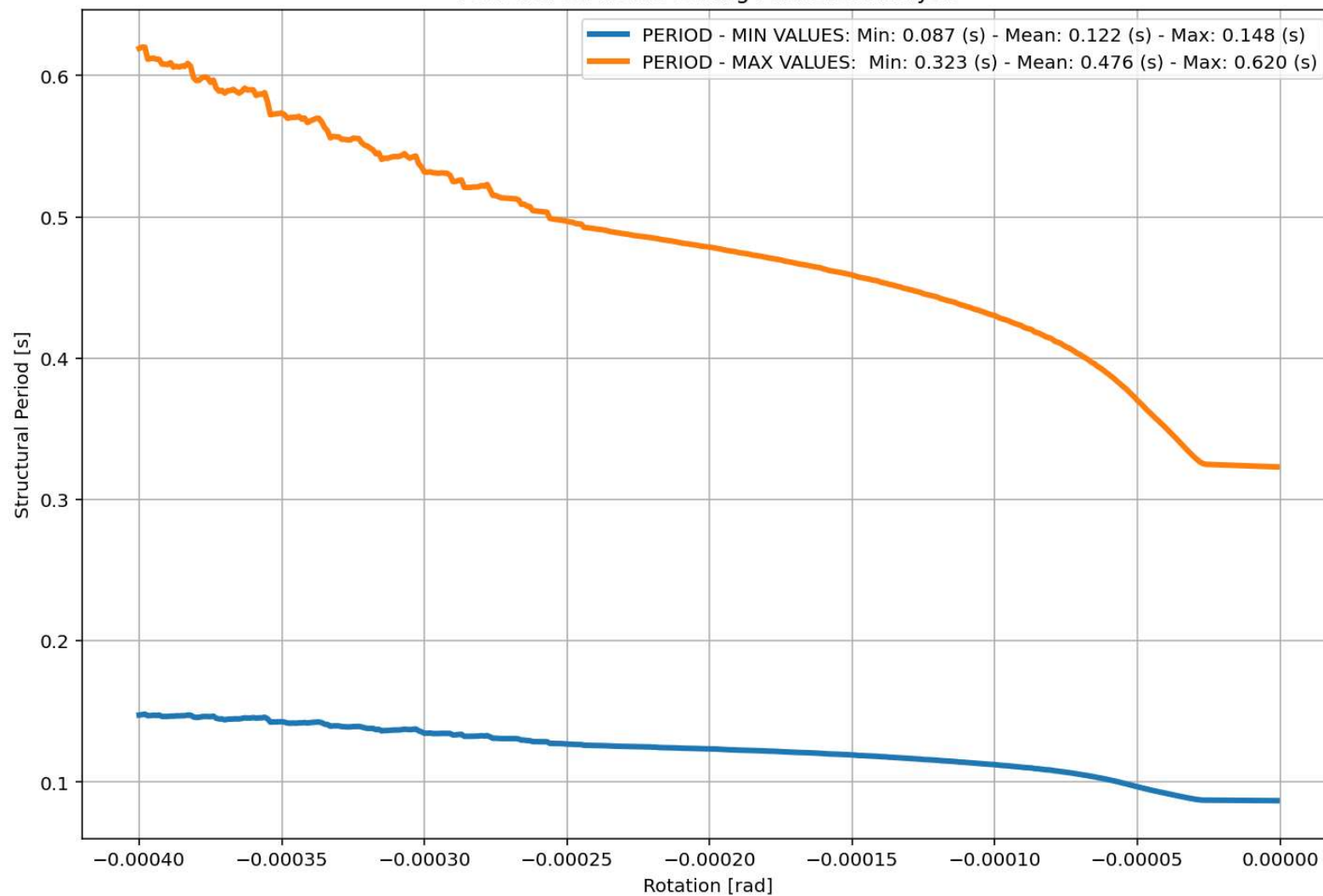
Moment and Rotation of Structure During Pushover Analysis

Python Console Files Help Variable Explorer Debugger Plots History

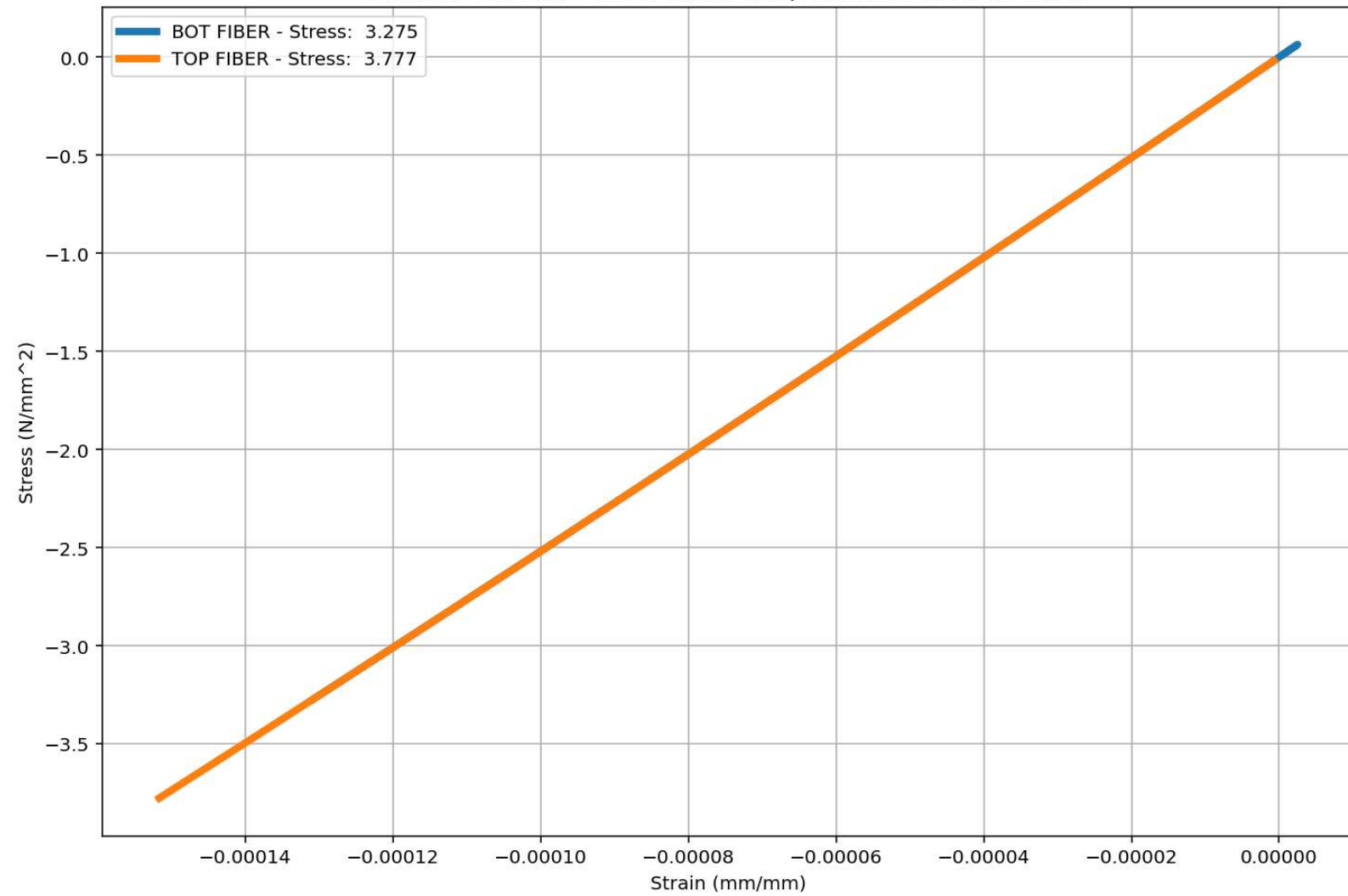
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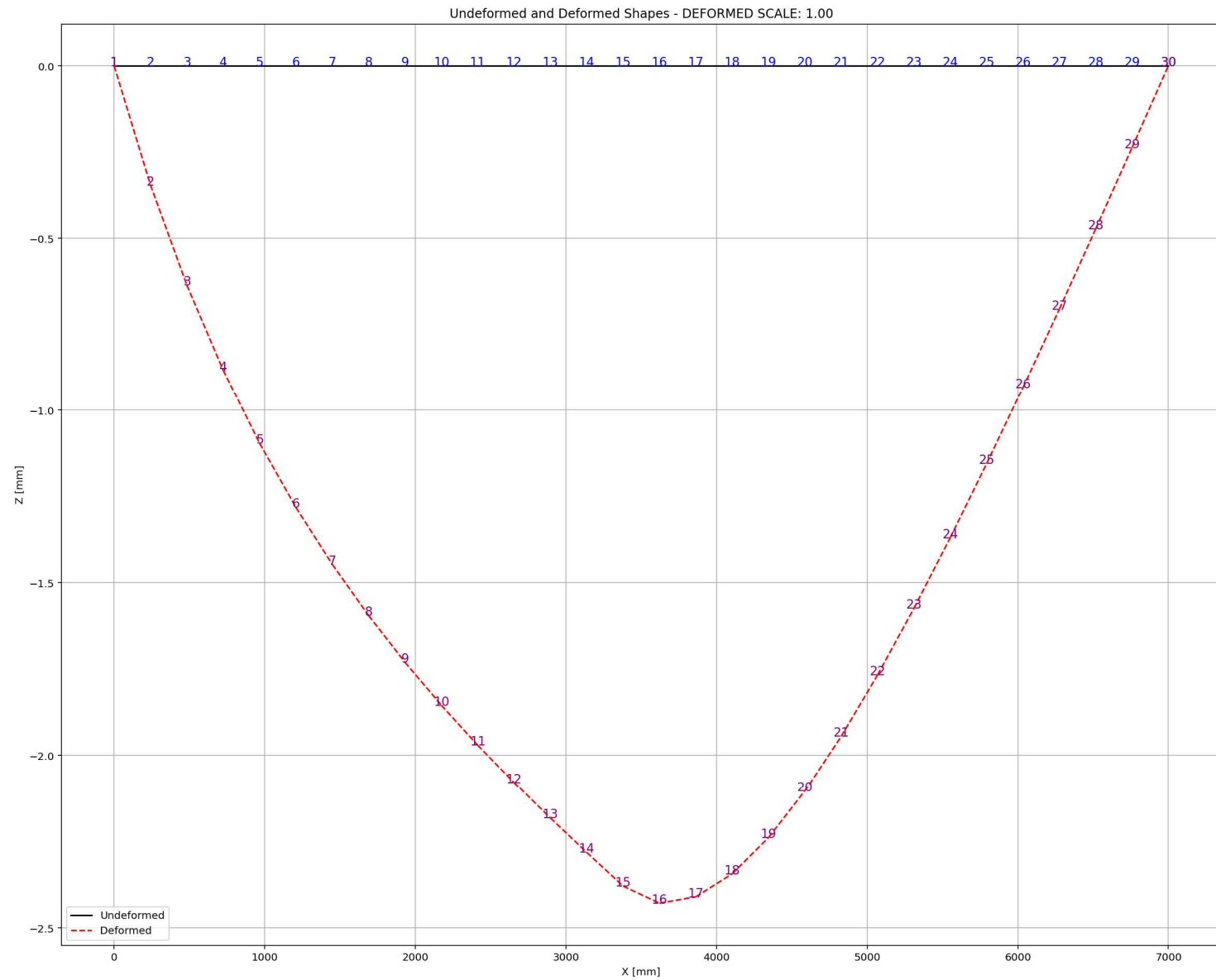


Period of Structure During Pushover Analysis

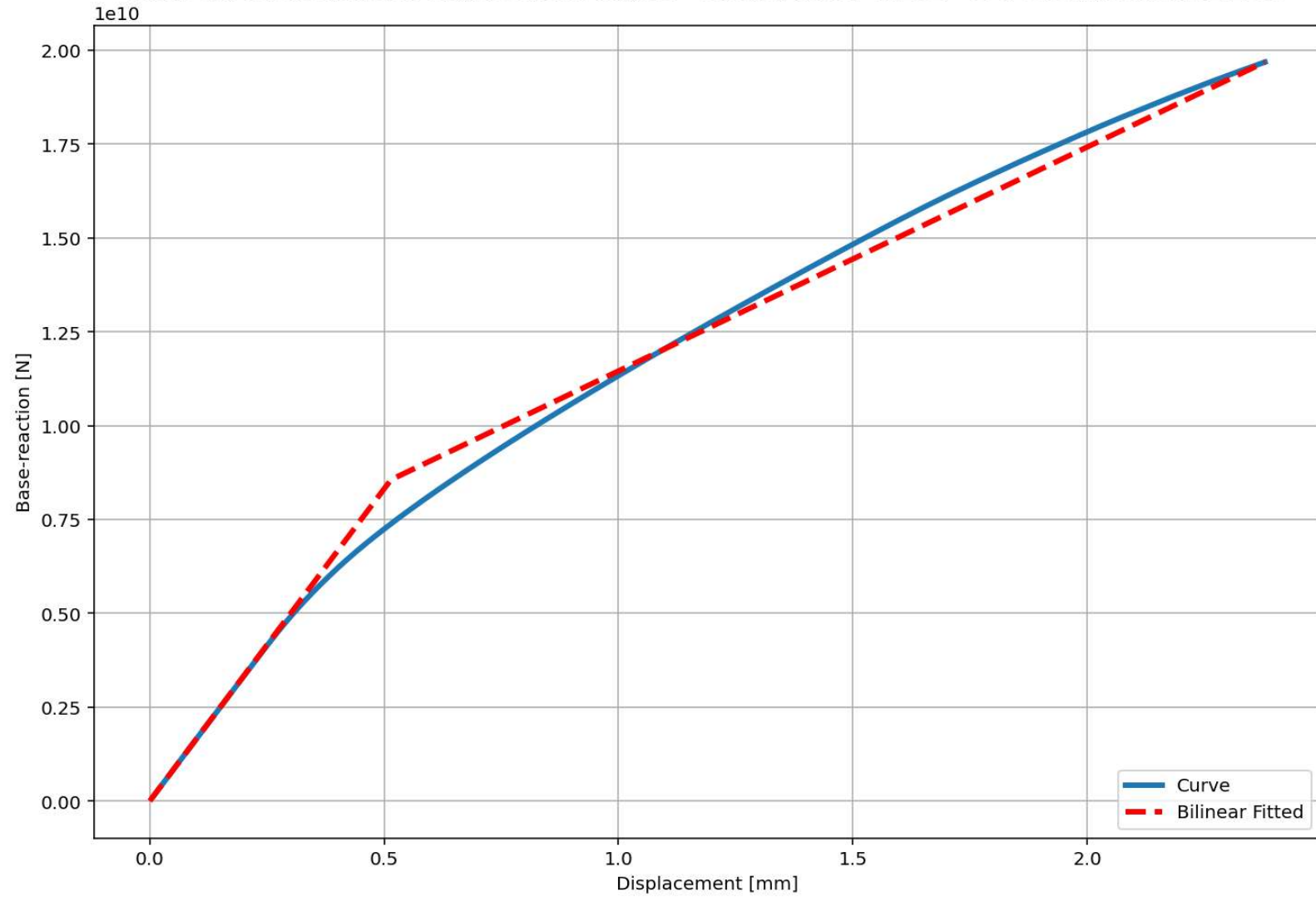


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve





Last Data of BaseReaction-Displacement Analysis - Ductility Ratio: 4.6232 - Over Strength Factor: 2.2998



+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]

[5.14623391e-01 8.55871071e+09]

[2.37919579e+00 1.96836283e+10]]

+=====+

+-----+

Structure Elastic Stiffness : 16631017660.17

Structure Plastic Stiffness : 8273227590.39

Structure Tangent Stiffness : 5966471217.87

Structure Ductility Ratio : 4.62

Structure Over Strength Factor: 2.30

+-----+

Over Strength Coefficient (Ω_0): 2.2998

Displacement Ductility Ratio (μ): 4.6232

Ductility Coefficient (R_μ): 4.6232

Structural Behavior Coefficient (R): 10.6326